Quantum-Inspired Cosmological Optimisation

A Novel Metaheuristic for the Maximum Cut Problem

Kritchanat Thanapiphatsiri

Department of Computer Engineering
Kasetsart University

2026-03-13

Outline

1. The Maximum Cut Problem . . 3
   - 1.1 Problem Definition 4
   - 1.2 Max-Cut Example 5
   - 1.3 Why Max-Cut? 6
2. The QICO Algorithm 7
   - 2.1 Motivation 8
   - 2.2 Algorithm Overview 9
   - 2.3 Gravitational Attraction . 10
   - 2.4 Quantum Tunnelling 11
   - 2.5 Dark Energy & Inflation . 12
     - 2.5.1 Dark Energy Perturbation 12
     - 2.5.2 Cosmic Inflation Restart 12
     - 2.5.1 Dark Energy Perturbation 12
     - 2.5.2 Cosmic Inflation Restart 12
   - 2.6 Gravitational Collapse . . . 13
   - 2.7 CMB Memory Archive . . 14
3. What Makes QICO Novel? . . 15
   - 3.1 Related Methods 16
   - 3.2 Key Innovations 17
4. Experimental Evaluation 18

Outline

4.1 Experimental Setup 19
4.2 Results: Small Graphs ($n = 20$) 20
4.3 Results: Med Graphs ($n = 50$) 21
4.4 Results: Large Graphs ($n = 100$) 22
4.5 Summary of Results 23
4.6 Strengths and Limitations 24
4.6.1 Strengths 24
4.6.2 Limitations 24

4.6.1 Strengths 24
4.6.2 Limitations 24

1. The Maximum Cut Problem

1.1 Problem Definition

Definition 1.1.1 Given a weighted undirected graph $G = (V, E, w)$, the **Maximum Cut** (Max-Cut) problem asks for a partition of V into two disjoint sets S and $\overline{S} = V \setminus S$ that maximises the total weight of edges crossing the partition:

$$\mathrm{Cut}(S) = \sum_{\substack{(u,v)\in E \\ u\in S, v\in \overline{S}}} w(u, v)$$

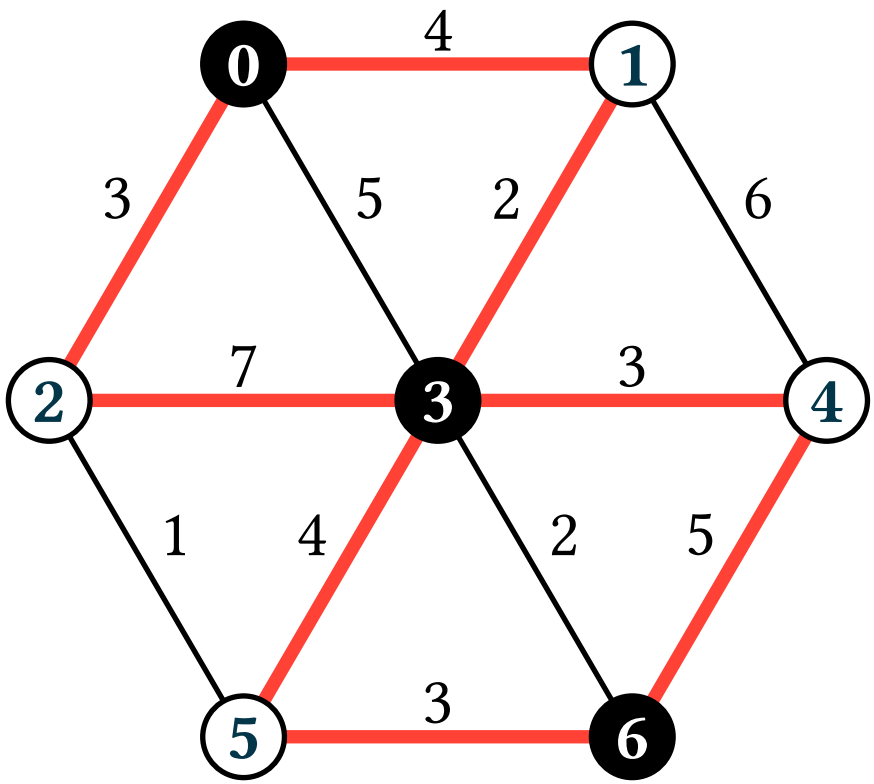
1.1 Problem Definition

Definition 1.1.1 Given a weighted undirected graph $G = (V, E, w)$, the **Maximum Cut** (Max-Cut) problem asks for a partition of V into two disjoint sets S and $\overline{S} = V \setminus S$ that maximises the total weight of edges crossing the partition:

$$\mathrm{Cut}(S) = \sum_{\substack{(u,v)\in E \\ u\in S, v\in \overline{S}}} w(u, v)$$

- **NP-hard** – one of Karp's 21 NP-complete problems (decision version)
- Best known polynomial approximation: **Goemans–Williamson** (0.878-approximation via SDP)
- Canonical benchmark for **QAOA** (Quantum Approximate Optimisation Algorithm)

1.2 Max-Cut Example



Partition: $S = \{0, 3, 6\}$

Cut edges (red):

$(0,1) : 4, \quad (0,2) : 3$
$(1,3) : 2, \quad (2,3) : 7$
$(3,4) : 3, \quad (3,5) : 4$
$(4,6) : 5, \quad (5,6) : 3$

$$\textbf{Cut value} = 2(4) + 3(3) + 2 + 7 + 5$$
$$= 31$$

1.3 Why Max-Cut?

Graph Theory Fundamental graph partitioning problem with applications in VLSI design, statistical physics, and network analysis

Quantum Computing The problem naturally maps to an **Ising Hamiltonian**:

$$H = \sum_{(i,j)\in E} w_{ij} \cdot \sigma_i^z \sigma_j^z$$

making it the primary benchmark for quantum optimisation (QAOA, quantum annealing)

Difficulty NP-hard even for unweighted graphs; no PTAS unless P = NP

2. The QICO Algorithm

2.1 Motivation

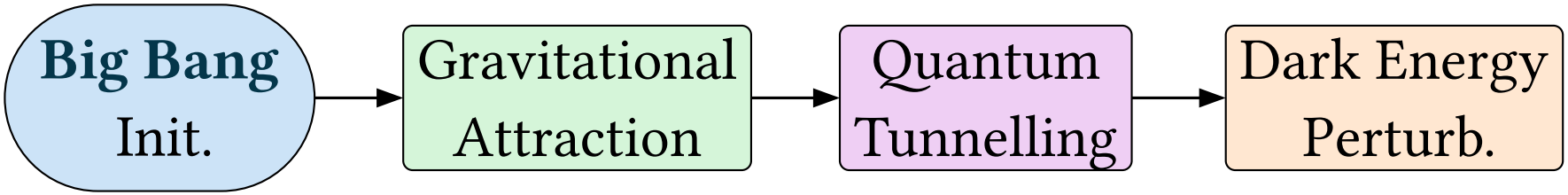
The evolution of the universe naturally balances **exploration** and **exploitation**:

2.1 Motivation

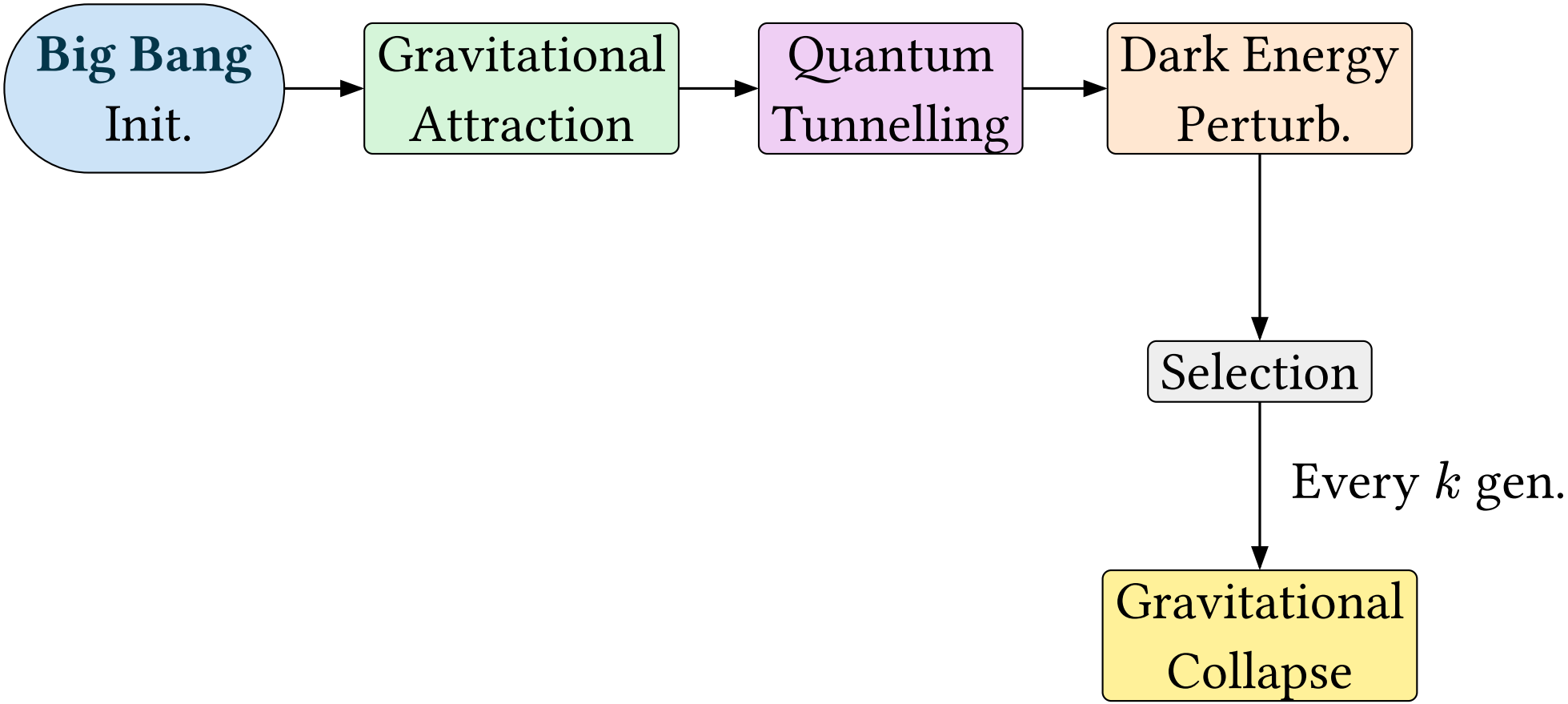
The evolution of the universe naturally balances **exploration** and **exploitation**:

Cosmological Phenomenon	**Optimisation Analogue**
Big Bang	Random initialisation
Cosmic expansion (Hubble flow)	Exploration of search space
Gravitational attraction	Exploitation towards good regions
Quantum tunnelling	Escaping local optima
Dark energy	Preventing premature convergence
Structure formation	Local search refinement
CMB radiation	Memory of best configurations

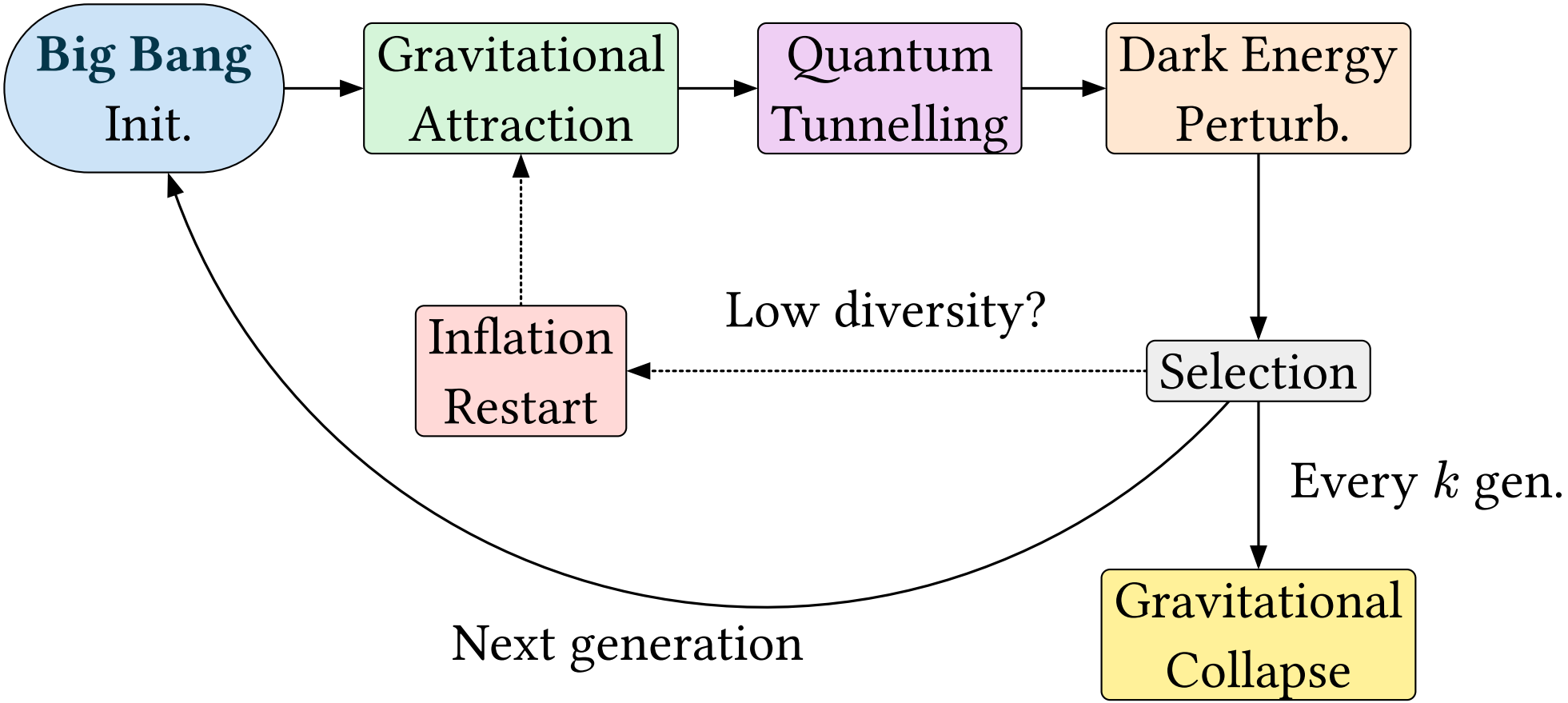
2.2 Algorithm Overview



2.2 Algorithm Overview



2.2 Algorithm Overview



2.3 Gravitational Attraction

Each solution x_i has a **mass** proportional to its fitness:

$$m_i = \frac{f(x_i) - f_{\min}}{f_{\max} - f_{\min}}$$

2.3 Gravitational Attraction

Each solution x_i has a **mass** proportional to its fitness:

$$m_i = \frac{f(x_i) - f_{\min}}{f_{\max} - f_{\min}}$$

For each solution, select an **attractor** x_j from the top-k solutions with probability $\propto m_j$.

2.3 Gravitational Attraction

Each solution x_i has a **mass** proportional to its fitness:

$$m_i = \frac{f(x_i) - f_{\min}}{f_{\max} - f_{\min}}$$

For each solution, select an **attractor** x_j from the top-k solutions with probability $\propto m_j$.

For each bit b where $x_i^{(b)} \neq x_j^{(b)}$, flip $x_i^{(b)}$ towards the attractor with probability:

$$P_{\text{grav}} = (1 - H(t)) \cdot m_j$$

where $H(t) = H_0 \cdot \alpha^t$ is the **Hubble parameter** that decays over time (from exploration to exploitation).

2.4 Quantum Tunnelling

Solutions can **tunnel through energy barriers** to escape local optima.

2.4 Quantum Tunnelling

Solutions can **tunnel through energy barriers** to escape local optima.

The tunnelling probability depends on the **energy barrier** and **quantum temperature**:

$$\Delta E_i = \frac{f^* - f(x_i)}{f^*}, \qquad T_q(t) = T_0 \cdot \beta^t$$

$$P_{\text{tunnel}} = \exp\left(-\Delta \frac{E_i}{T_q}(t)\right)$$

2.4 Quantum Tunnelling

Solutions can **tunnel through energy barriers** to escape local optima.

The tunnelling probability depends on the **energy barrier** and **quantum temperature**:

$$\Delta E_i = \frac{f^* - f(x_i)}{f^*}, \qquad T_q(t) = T_0 \cdot \beta^t$$

$$P_{\text{tunnel}} = \exp\left(-\Delta \frac{E_i}{T_q}(t)\right)$$

When tunnelling occurs, n_{flip} random bits are flipped, where:

$$n_{\text{flip}} \sim \text{Geometric}(p = \max(0.3, 1 - H(t)))$$

2.4 Quantum Tunnelling

Solutions can **tunnel through energy barriers** to escape local optima.

The tunnelling probability depends on the **energy barrier** and **quantum temperature**:

$$\Delta E_i = \frac{f^* - f(x_i)}{f^*}, \qquad T_q(t) = T_0 \cdot \beta^t$$

$$P_{\text{tunnel}} = \exp\left(-\Delta \frac{E_i}{T_q}(t)\right)$$

When tunnelling occurs, n_{flip} random bits are flipped, where:

$$n_{\text{flip}} \sim \text{Geometric}(p = \max(0.3, 1 - H(t)))$$

Key insight: better solutions (low ΔE) tunnel more often but flip fewer bits, while poor solutions occasionally make large jumps.

2.5 Dark Energy & Inflation

2.5.1 Dark Energy Perturbation

Each bit is randomly flipped with probability $p_{\mathrm{DE}} \cdot H(t)$, where p_{DE} is a small constant.

- Prevents premature convergence by maintaining diversity
- Effect diminishes naturally as $H(t)$ decays

2.5 Dark Energy & Inflation

2.5.1 Dark Energy Perturbation

Each bit is randomly flipped with probability $p_{\mathrm{DE}} \cdot H(t)$, where p_{DE} is a small constant.

- Prevents premature convergence by maintaining diversity
- Effect diminishes naturally as $H(t)$ decays

2.5.2 Cosmic Inflation Restart

When the population **diversity** (mean pairwise Hamming distance) drops below a threshold:

- Retain the **elite** (top 50% by fitness)
- Replace the **bottom 50%** with fresh random solutions

This is analogous to cosmic inflation – a rapid expansion event that re-introduces diversity into a stagnating universe.

2.6 Gravitational Collapse

Periodically (every k generations), the **top-K** elite solutions undergo **local search**:

2.6 Gravitational Collapse

Periodically (every k generations), the **top-K** elite solutions undergo **local search**:

Gravitational Collapse (1-flip local search)

For each elite solution x:

repeat:

For each vertex $v \in V$ (random order):

Compute $\Delta = \sum_{u \in N(v)} w(v,u) \cdot \begin{cases} +1 & \text{if } x_u = x_v \\ -1 & \text{if } x_u \neq x_v \end{cases}$

If $\Delta > 0$: flip x_v

until no improvement found

2.6 Gravitational Collapse

Periodically (every k generations), the **top-K** elite solutions undergo **local search**:

Gravitational Collapse (1-flip local search)

For each elite solution x:

repeat:

For each vertex $v \in V$ (random order):

Compute $\Delta = \sum_{u \in N(v)} w(v,u) \cdot \begin{cases} +1 & \text{if } x_u = x_v \\ -1 & \text{if } x_u \neq x_v \end{cases}$

If $\Delta > 0$: flip x_v

until no improvement found

This is $O(|E|)$ per pass and converges to a **local optimum** with respect to single-vertex moves.

2.7 CMB Memory Archive

The **Cosmic Microwave Background** (CMB) memory preserves the best solutions discovered throughout the search:

- Maintains an archive of the top-M historically best solutions
- Every K generations, a random archived solution replaces the **worst** individual in the population

2.7 CMB Memory Archive

The **Cosmic Microwave Background** (CMB) memory preserves the best solutions discovered throughout the search:

- Maintains an archive of the top-M historically best solutions
- Every K generations, a random archived solution replaces the **worst** individual in the population

Purpose: prevents the loss of high-quality genetic material during cosmic inflation restarts or stagnation periods.

This is analogous to the CMB – a relic radiation from the early universe that carries information about initial conditions.

3. What Makes QICO Novel?

3.1 Related Methods

Feature	**GSA**	**BB-BC**	**QICO**
Domain	Continuous	Continuous	**Combinatorial**
Quantum tunnelling	No	No	**Adaptive**
Diversity restart	No	Big Crunch	**Inflation trigger**
Local search	No	No	**Grav. collapse**
Solution archive	No	No	**CMB memory**
Parameter adaptation	Fixed G	Fixed	**Hubble decay**

GSA Gravitational Search Algorithm (Rashedi et al., 2009)

BB-BC Big Bang–Big Crunch (Erol & Eksin, 2006)

3.2 Key Innovations

1. **Adaptive quantum tunnelling** with energy-barrier-aware probability
   - Solutions near the optimum tunnel frequently (fine-tuning)
   - Distant solutions occasionally make large jumps (exploration)

3.2 Key Innovations

1. **Adaptive quantum tunnelling** with energy-barrier-aware probability
   - Solutions near the optimum tunnel frequently (fine-tuning)
   - Distant solutions occasionally make large jumps (exploration)
2. **Hubble parameter** for smooth exploration $\rightarrow$ exploitation transition
   - Controls gravitational force, tunnelling magnitude, and dark energy simultaneously
   - Single parameter governs the entire search dynamics

3.2 Key Innovations

1. **Adaptive quantum tunnelling** with energy-barrier-aware probability
   - Solutions near the optimum tunnel frequently (fine-tuning)
   - Distant solutions occasionally make large jumps (exploration)
2. **Hubble parameter** for smooth exploration $\rightarrow$ exploitation transition
   - Controls gravitational force, tunnelling magnitude, and dark energy simultaneously
   - Single parameter governs the entire search dynamics
3. **Diversity-triggered cosmic inflation**
   - Automatic detection of premature convergence
   - Targeted re-diversification preserving elite knowledge

3.2 Key Innovations

1. **Adaptive quantum tunnelling** with energy-barrier-aware probability
   - Solutions near the optimum tunnel frequently (fine-tuning)
   - Distant solutions occasionally make large jumps (exploration)
2. **Hubble parameter** for smooth exploration $\rightarrow$ exploitation transition
   - Controls gravitational force, tunnelling magnitude, and dark energy simultaneously
   - Single parameter governs the entire search dynamics
3. **Diversity-triggered cosmic inflation**
   - Automatic detection of premature convergence
   - Targeted re-diversification preserving elite knowledge
4. **Integrated local search** (gravitational collapse)
   - Efficient $O(|E|)$ delta evaluation
   - Applied selectively to top solutions only

4. Experimental Evaluation

4.1 Experimental Setup

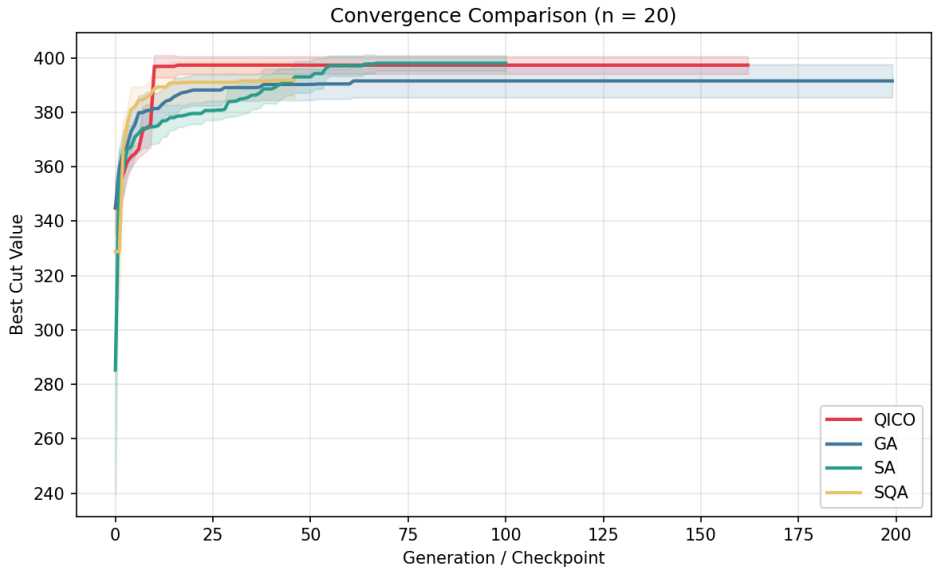
Problem instances: Random Erdos–Renyi weighted graphs $G(n, p)$ with weights $w \in [1, 10]$

n	Density	$\lvert E \rvert$	Budget
20	0.5	≈ 105	10 000
50	0.4	≈ 486	30 000
100	0.3	≈ 1523	60 000

Algorithms compared (10 independent trials each):

- **QICO** – proposed method (pop. size = 50)
- **GA** – Genetic algorithm (tournament, uniform crossover, bit-flip mutation)
- **SA** – Simulated annealing (geometric cooling)
- **SQA** – Simulated quantum annealing (path-integral Monte Carlo, 8 replicas)

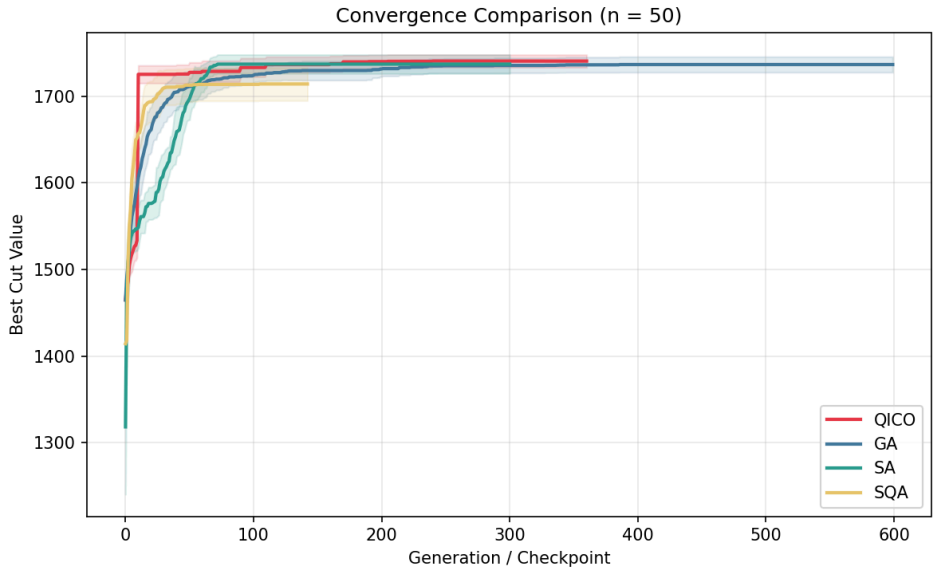
4.2 Results: Small Graphs ($n = 20$)



Algorithm	Mean	Std
QICO	**397.3**	3.2
GA	391.5	6.1
SA	398.0	2.8
SQA	391.8	6.6

- All methods find the optimum (399.4) at least once
- SA and QICO are most consistent
- GA and SQA show higher variance

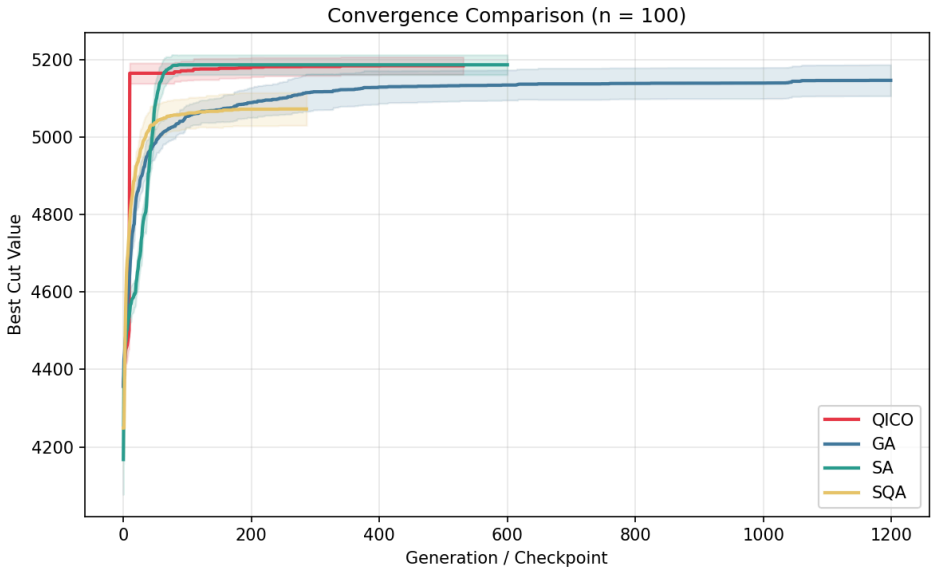
4.3 Results: Med Graphs ($n = 50$)



Algorithm	Mean	Std
QICO	**1740.7**	**7.3**
GA	1736.7	9.0
SA	1737.2	11.0
SQA	1714.3	19.1

- QICO achieves the **best mean** and **lowest variance**
- SQA falls behind due to limited per-replica budget
- QICO's local search gives it an edge

4.4 Results: Large Graphs ($n = 100$)



Algorithm	Mean	Std
QICO	**5184.6**	22.5
GA	5146.8	40.3
SA	5187.0	25.5
SQA	5072.8	42.4

- QICO and SA are close; both significantly outperform GA and SQA
- QICO's strength: **population diversity** + **local search**
- SQA struggles with the evaluation budget split across replicas

4.5 Summary of Results

Metric	$n = 20$	$n = 50$	$n = 100$
Best mean	SA (398.0)	**QICO (1740.7)**	SA (5187.0)
Lowest std	SA (2.8)	**QICO (7.3)**	**QICO (22.5)**
2nd best mean	**QICO (397.3)**	SA (1737.2)	**QICO (5184.6)**

- QICO is **consistently among the top 2** across all problem sizes
- Most reliable (lowest or near-lowest variance)
- Combines the population diversity of GA with the local refinement of SA
- Significantly outperforms SQA under fixed evaluation budgets

4.6 Strengths and Limitations

4.6.1 Strengths

- **Robust**: low variance across independent runs
- **Adaptive**: Hubble parameter smoothly transitions search behaviour
- **Modular**: each cosmological phase addresses a specific optimisation need
- Quantum tunnelling provides **principled** barrier crossing

4.6 Strengths and Limitations

4.6.1 Strengths

- **Robust**: low variance across independent runs
- **Adaptive**: Hubble parameter smoothly transitions search behaviour
- **Modular**: each cosmological phase addresses a specific optimisation need
- Quantum tunnelling provides **principled** barrier crossing

4.6.2 Limitations

- More **hyperparameters** than SA (though defaults work well)
- Local search dominates on small instances (the metaheuristic overhead is less impactful)
- **Population-based**: higher per-generation cost than single-solution methods

Thank you!

Any questions?